

MBS-007- Cognitive Convergence

A Trigger-Centered Interpretation of Why Minds Become Similar

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Repository: <https://github.com/sizhet/MET-Brain-Structure-MBS>

Introduction

One of the most remarkable observations in biology is that cognition often converges.

Different individuals frequently develop similar understandings of their environments.

Different animals often exhibit surprisingly similar behaviors.

Different societies repeatedly rediscover similar solutions.

Across evolutionary history, many cognitive structures appear again and again.

Why does this happen?

Traditional explanations typically emphasize:

- Genetics
- Natural selection
- Learning
- Environment
- Culture
- Logic

These explanations are important and undoubtedly contribute to convergence.

However, the MET Brain Structure (MBS) project proposes an additional possibility:

Cognitive convergence may emerge from Trigger Convergence.

This perspective does not replace existing explanations.

Instead, it attempts to identify a deeper structural mechanism that may help explain why convergence appears so frequently throughout nature.

The Traditional View

A common explanation is straightforward.

Different organisms face similar environments.

Similar environments favor similar behaviors.

Over time, similar behaviors produce similar cognition.

This explanation is largely correct.

However, it leaves open an important question:

What structural mechanism transforms environmental similarity into cognitive similarity?

The MBS perspective proposes that trigger systems may play a central role.

Why Triggers Matter

Every cognitive system must decide:

What deserves attention?

What requires action?

What should be ignored?

This decision cannot be postponed.

It must occur continuously.

Trigger systems perform this function.

They determine significance.

They determine urgency.

They determine relevance.

Before retrieval.

Before planning.

Before reasoning.

Something must first become important.

This is the role of triggering.

The Trigger Bottleneck

A useful thought experiment is to imagine two organisms possessing identical knowledge but different trigger systems.

Their behavior would likely differ dramatically.

One might notice danger immediately.

The other might ignore it.

One might react to social signals.

The other might not.

One might pursue opportunities aggressively.

The other might remain passive.

Despite sharing knowledge, their cognition would diverge.

Now consider the opposite situation.

Suppose two organisms possess different experiences but similar trigger systems.

They may repeatedly focus on similar categories of events.

Over time, this repeated attention can drive convergence.

This suggests that triggers may occupy a uniquely powerful position within cognition.

Trigger Categories

Many trigger categories appear repeatedly across biological systems.

Examples include:

- Predators
- Food
- Offspring
- Mates
- Territory
- Pain
- Social status
- Shelter
- Novelty
- Opportunity

These categories emerge because they repeatedly influence survival and reproduction.

As a result, they become evolutionarily significant.

Different organisms may therefore develop remarkably similar trigger priorities.

From Trigger Convergence to Cognitive Convergence

The MBS hypothesis proposes the following sequence:

Similar Environment

↓

Similar Survival Pressures

↓

Similar Trigger Categories

↓

Similar Retrieval Requirements

↓

Similar Functional Tunnels

↓

Similar Scene Structures

↓

Similar Cognitive Patterns

In this framework, cognition converges because triggering converges.

The trigger system acts as a structural funnel.

Many possible experiences become filtered through similar priorities.

Over time, cognition begins to resemble itself across individuals and species.

Retrieval Convergence

Once trigger categories become similar, retrieval systems face similar demands.

For example:

A predator-related trigger requires:

- Threat recognition
- Escape memories
- Environmental knowledge
- Behavioral pathways

Across generations, retrieval structures capable of supporting these needs are repeatedly reinforced.

This creates retrieval convergence.

Different minds become organized around similar retrieval challenges.

Functional Tunnel Convergence

The same principle extends to Function Tunnels.

Many organisms repeatedly encounter:

Threat

↓

Escape

Food

↓

Acquisition

Offspring

↓

Protection

Territory

↓

Defense

These recurring requirements encourage the emergence of similar functional pathways.

Although implementation details differ, the structural patterns often remain recognizable.

Function Tunnel convergence therefore becomes a natural consequence of

Trigger convergence.

Scene Convergence

The Embodied Scene Runtime perspective introduces another level of convergence.

Triggers activate scenes.

Scenes provide context.

Context shapes experience.

Over time, organisms repeatedly exposed to similar triggers may develop similar scene structures.

Examples include:

- Predator scenes
- Hunting scenes
- Shelter scenes
- Social scenes
- Parenting scenes

The resulting cognitive landscape begins to exhibit recurring patterns.

Why Dreams Matter

Dreaming may play an important role in this process.

Dreams repeatedly activate:

- Triggers
- Retrieval pathways
- Functional tunnels
- Scene structures

In doing so, they may reinforce convergence.

Different organisms living under similar conditions may repeatedly replay similar structural themes.

Dreaming therefore becomes not only a maintenance mechanism but also a potential convergence mechanism.

Human Cognition

The same principles may extend into human cognition.

Humans possess language, culture, and abstract reasoning.

Yet many fundamental trigger categories remain remarkably stable.

Threat.

Opportunity.

Social belonging.

Status.

Curiosity.

Safety.

Attachment.

Achievement.

These recurring trigger categories continue to shape retrieval, attention, memory, planning, and learning.

Even highly abstract thought may ultimately be influenced by deep triggering structures inherited from much older cognitive architectures.

Civilization-Level Convergence

The idea may extend even further.

Different civilizations often rediscover:

Agriculture.

Trade.

Governance.

Mathematics.

Navigation.

Engineering.

Communication systems.

Traditional explanations emphasize practical necessity.

The MBS perspective agrees but adds a structural interpretation.

Civilizations may converge because large populations share similar trigger categories.

Shared triggers produce shared retrieval pressures.

Shared retrieval pressures encourage similar functional pathways.

Over time, civilizations may converge structurally even when separated geographically.

Trigger Systems as Cognitive Attractors

A useful way to think about triggers is as attractors.

They continuously pull cognition toward particular regions of experience.

Danger pulls attention.

Food pulls exploration.

Social signals pull interpretation.

Goals pull planning.

These attractors shape the long-term trajectory of cognitive development.

The resulting convergence may therefore emerge naturally without requiring centralized design.

Relationship to Modern Neuroscience

Modern neuroscience has identified many systems involved in:

- Salience detection
- Attention
- Reward processing
- Threat processing
- Motivation
- Emotional regulation

The Trigger Convergence hypothesis does not replace these findings. Instead, it offers a structural interpretation.

The proposal is that convergence may emerge because many organisms repeatedly organize cognition around similar trigger structures.

Future neuroscience research will determine how well this interpretation corresponds to biological reality.

Implications for Artificial Intelligence

The Trigger Convergence perspective may also have implications for AI. Many current AI systems focus primarily on knowledge representation and reasoning.

The MBS framework suggests that triggering may be equally important.

Future intelligent systems may require:

- Trigger architectures
- Priority systems
- Retrieval governance
- Functional tunnel activation
- Scene-centered runtime structures

Without triggers, intelligence lacks direction.

Without direction, convergence becomes difficult.

Understanding trigger systems may therefore become a major challenge for future AI research.

Conclusion

Cognitive convergence is one of the most pervasive phenomena in nature.

Individuals converge.

Species converge.

Cultures converge.

Civilizations converge.

The MET Brain Structure project proposes that a significant portion of this convergence may originate from Trigger Convergence.

Similar environments create similar survival pressures.

Similar survival pressures create similar triggers.

Similar triggers shape retrieval.

Retrieval shapes tunnels.

Tunnels shape scenes.

Scenes shape cognition.

From this perspective, minds become similar not because they begin with identical knowledge, but because they repeatedly organize themselves around similar sources of significance.

In the language of MBS:

Environment shapes triggers.

Triggers shape structure.

Structure shapes cognition.

And cognition gradually converges.